ML Based Student Advisor

Predicting Academic Success with Machine Learning

Presented By

Mohammad Injamul Haque (ID – 231-115-019)
Riajul Hasan (ID – 221-115-112)
Arup Ratan Dey (ID – 231-115-013)

Implemented by: Python, Scikit-learn, GroqCloud AI

Problem Statement

The Challenges:

- High student failure rates
- Late identification of at-risk students
- Generic academic advice
- No personalized intervention

Our Solutions:

- Early prediction of final grades
- AI-powered personalized suggestions
- Risk level classification
- Data-driven insights for educators

Objectives

Project Goals

1

1. Predictive Analytics
   - Forecast student final grades (G3)
   - Identify at-risk students early

2

2. Personalized Recommendations
   - AI-generated study strategies
   - Customized improvement plans

3

3. Educator Support
   - Data-driven insights
   - Resource allocation guidance

4

4. Accessible Interface
   - Easy-to-use analysis tool
   - PDF report generation

Dataset Overview

Student Performance Dataset

Source Link : UC Irvine

Dataset Details

1	**Subjects**	Mathematics & Portuguese
2	**Students**	1,044 total
3	**Features**	33 variables per student
4	**Periods**	G1, G2, G3 (Three academic periods)

Dataset Overview

Student Performance Dataset

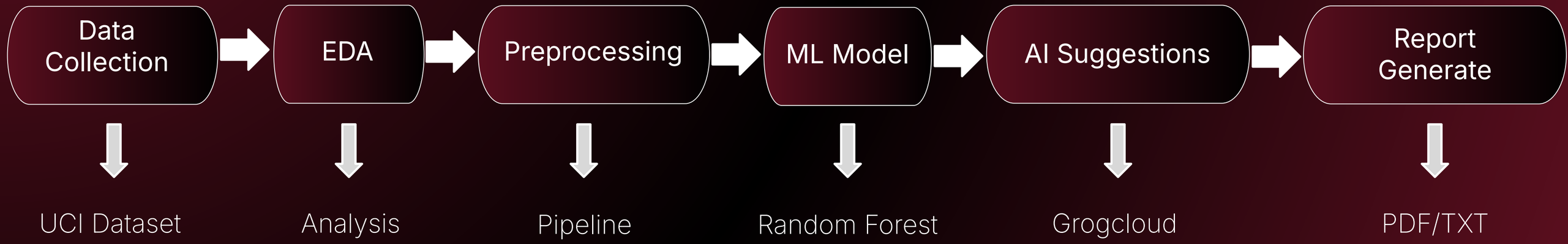
Source Link : UC Irvine

Key Features

1. **Demographic** — Age, Gender, Address
2. **Academic** — Previous Grades, Study Time, Absences
3. **Social** — Family Relations, Extra-curricular Activities
4. **Life-style** — Health-status, Alcohol consumption

Methodology

System Architecture



Exploratory Data Analysis

- Key Findings

Data Insights

1

Grade Distribution:

- Average G3: 11.34/20
- Pass Rate: 61.5% (≥10/20)
- Distribution: Approximately normal

2

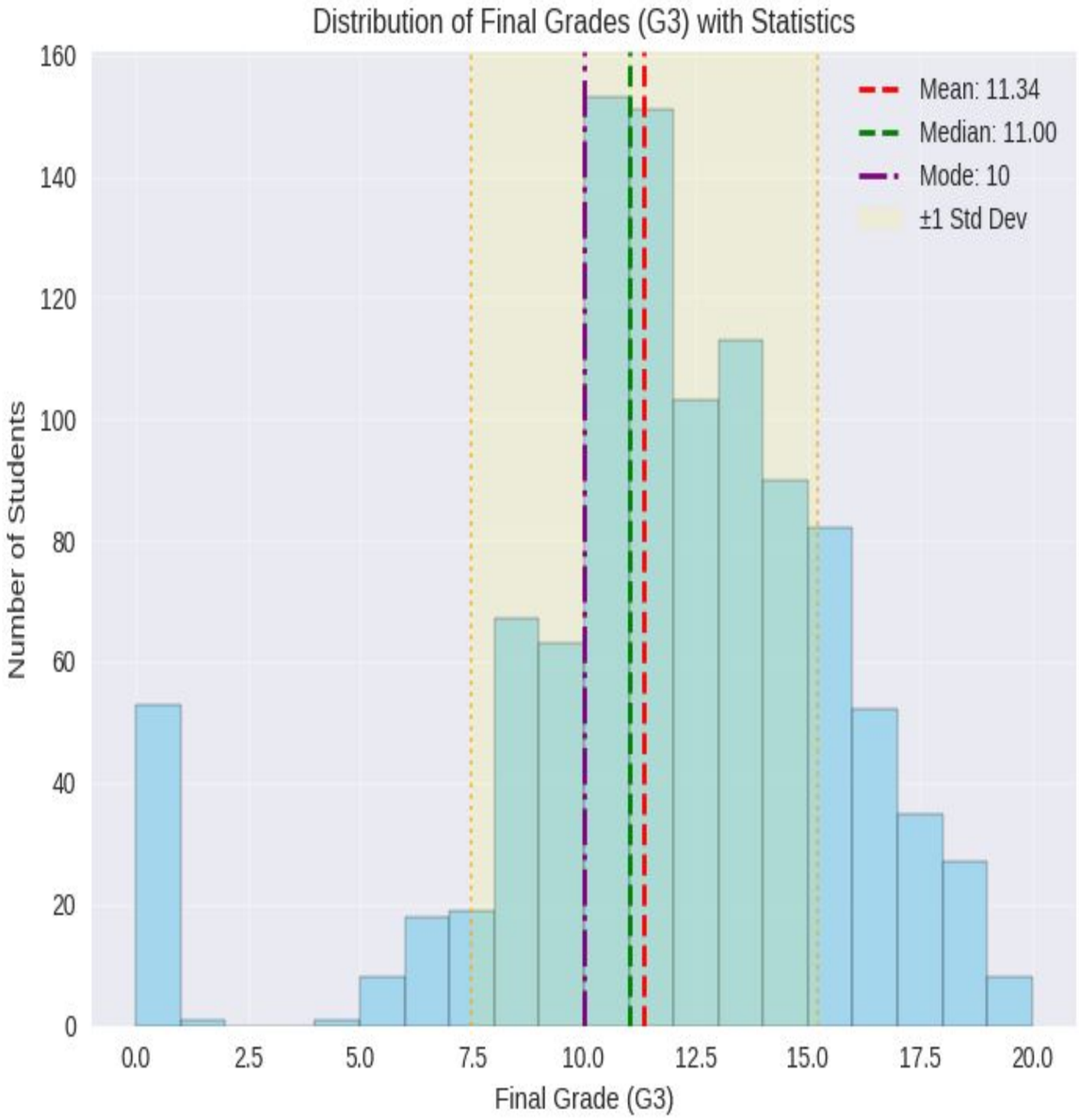
Study Time Impact:

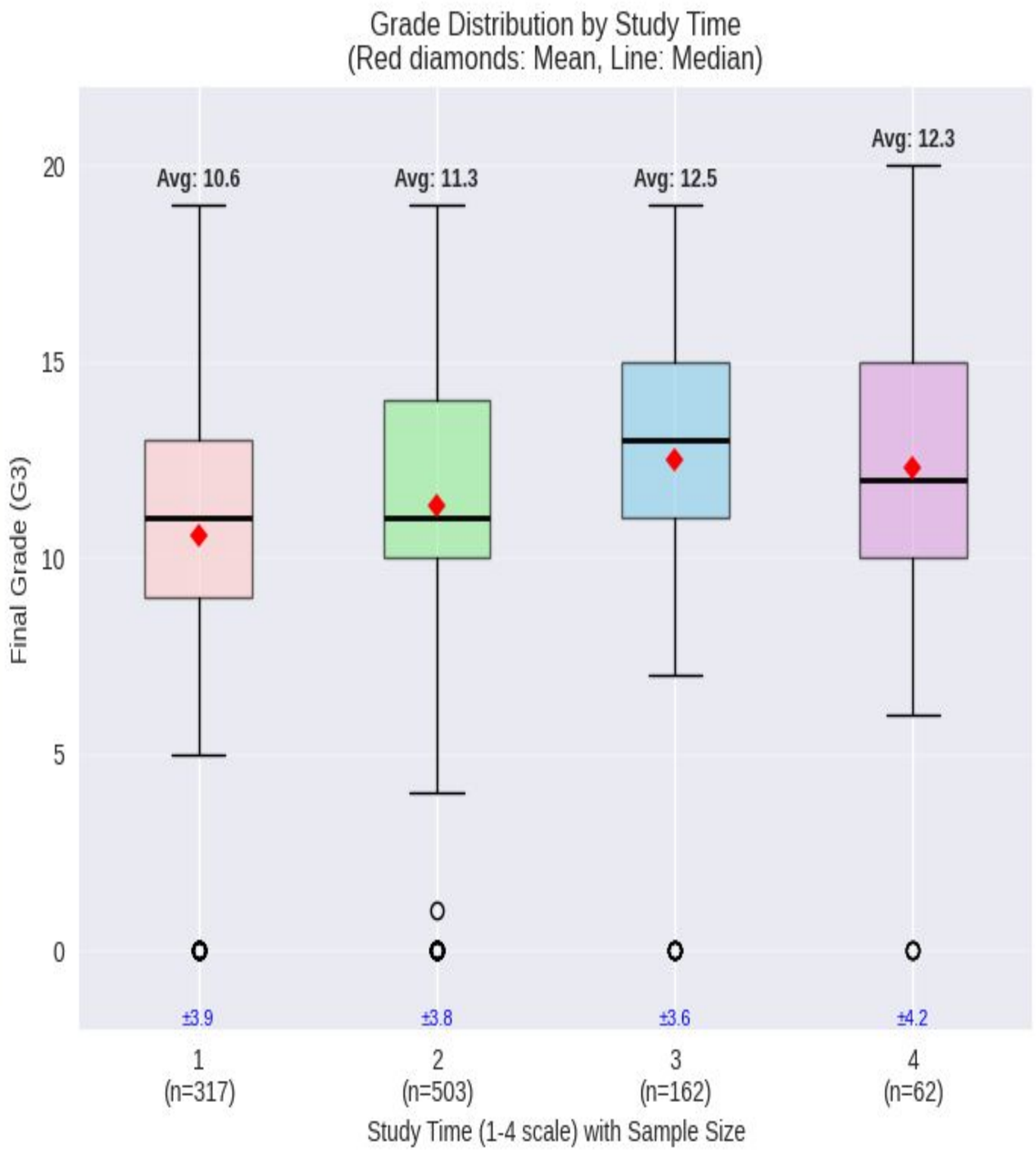
- Optimal: StudyTime 3 (5-10 hours/week)
- Diminishing returns: StudyTime 4 (>10 hours)
- Quality > Quantity

3

Key Correlations:

- G2 → G3: 0.91 (Strongest predictor)
- StudyTime → G3: 0.35 (Moderate)
- Absences → G3: -0.24 (Negative)


Distribution of Final Grades (G3) with Statistics
Mean: 11.34
Median: 11.00
Mode: 10
±1 Std Dev
Number of Students
Final Grade (G3)
0
20
40
60
80
100
120
140
160
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Grade Distribution by Study Time
(Red diamonds: Mean, Line: Median)
Avg: 10.6
Avg: 11.3
Avg: 12.5
Avg: 12.3
±3.9
±3.8
±3.6
±4.2
Final Grade (G3)
0
5
10
15
20
1
(n=317)
2
(n=503)
3
(n=162)
4
(n=62)
Study Time (1-4 scale) with Sample Size

Data Preprocessing Pipeline

Data Preparation Steps

1

Feature Separation:
- Target: G3 (Final Grade)
- Features: 32 variables

2

Data Cleaning:
- Handle missing values
- Numeric: Median imputation
- Categorical: Mode imputation

3

Feature Engineering:
- Numeric: Standard Scaling (mean=0, std=1)
- Categorical: One-Hot Encoding

4

Pipeline Creation:
- Consistent processing
- Prevents data leakage

Machine Learning Models

Regression Implementation

1

Regression Model (Grades):

- Algorithm: Random Forest Regressor
- Predicts: Final grade (G3) 0-20 scale
- Performance: R^2 = 0.81, RMSE = 1.71

2

Classification Model (Risk):

- Algorithm: Random Forest Classifier
- Categories: High/Medium/Low Risk
- Thresholds: ≤8, 9-12, ≥13

3

Feature Importance:

- Top predictor: G2 (84%)
- Second: Absences(5%)
- Third: Study Time(1%)

AI Powered Suggestion Engine

Functionality

1. Student Profile Analysis
2. Personalized Recommendations
3. Actionable Study Plans
4. Motivation & Support

Application Interface

Input Methods

1. Interactive Form
2. Sample Analysis
3. Main Application Menu

Application Interface

Output Features

1. Predicted Final Grade
2. Risk Level Assessment
3. AI generated suggestions
4. Printable PDF/TXT reports
5. Downloadable analysis

Application Interface

Main Menu Options

1. Analyze New Student
2. View Sample Analysis
3. View Saved Reports
4. View Model Performance
5. Exit

Result & Performance

Model Performance Metrics

1

Regression Model:

- R^2 Score: 0.81 (81% variance explained)
- RMSE: 1.707 points (out of 20)
- MAE: 0.963 points
- Accuracy: ±0.963 points on average

2

Classification Model:

- Accuracy: 83.3%
- High Risk Detection: 77.4%
- Low Risk Identification: 92.0%

3

AI Response Time:

- Average: 3-5 seconds per analysis
- Quality: Human-like, personalized advices

Sample Analysis Output

Student Analysis Example

Student Profile:

- Age: 17, Gender: Male
- Study Time: 2/4 (Low)
- Absences: 15 days
- Previous Grades: G1=6, G2=5

Predictions:

- Final Grade (G3): 6.3/20
- Risk Level: HIGH RISK
- Improvement Needed: 7.7 points

AI Recommendations

1. IMMEDIATE: Increase study time to 3/4
2. STRATEGY: Focus on weak subjects
3. LIFESTYLE: Reduce absences
4. RESOURCES: Seek teacher support
5. TIMELINE: Weekly progress checks

Technical Implementation

Tech Stack

1

Core Libraries:
- Data Processing: pandas, numpy
- Machine Learning: scikit-learn
- Visualization: matplotlib, seaborn
- API Integration: requests

2

Advanced Features:
- Pipeline Management: sklearn Pipeline
- Model Persistence: joblib
- Report Generation: reportlab, FPDF
- Error Handling: Comprehensive try-catch

3

Performance Optimizations:
- Efficient memory usage
- Fast API responses
- Parallel processing
- Caching mechanisms

Challenges & Solutions

Project Challenges

1. **Data Quality**
   - Challenge: Missing values, outliers
   - Solution: Robust imputation strategies
2. **Model Accuracy**
   - Challenge: Limited dataset size
   - Solution: Ensemble methods (Random Forest)
3. **AI Integration**
   - Challenge: API costs and limitations
   - Solution: Free tier optimization
4. **User Experience**
   - Challenge: Complex output presentation
   - Solution: Structured formatting
5. **Deployment**
   - Challenge: Dependency management
   - Solution: Google Colab compatibility

Educational Impact

Real World Example

1

For Students:
- Early warning system
- Personalized study plans
- Motivation and guidance
- Goal setting assistance

2

For Teachers:
- Identify at-risk students
- Data-driven interventions
- Resource optimization
- Parent communication tool

3

For Parents:
- Student progress monitoring
- Support strategy guidance
- Communication with teachers

Ethical Considerations

Responsible AI Implementation

1

Privacy & Security:
- Student data anonymization
- Secure API communication
- No personal identifier storage
- GDPR compliance considerations

2

Educational Ethics:
- Supplemental tool (not replacement)
- Teacher discretion maintained
- Positive reinforcement focus
- Growth mindset promotion

3

Social Impact:
- Reducing educational inequality
- Supporting diverse learners
- Promoting academic success
- Empowering educators

Conclusion

Key Takeways

1

Achievements:

1. Accurate grade prediction (81% accuracy)
2. Effective risk classification (83% accuracy)
3. Personalized AI recommendations
4. User-friendly interface
5. Comprehensive reporting

2

Innovation:

- Dual AI system (ML + Generative AI)
- Real-time analysis
- Actionable insights
- Scalable architecture

3

Future Vision:

- Expand to more educational contexts
- Integrate with school systems
- Continuous model improvement
- Global accessibility

Live Demo

Demonstration Time

1

What We'll Show:

1. Data loading and EDA
2. Model training process
3. Student analysis interface
4. AI suggestion generation
5. Report downloading

2

Key Features to Observe:

- Real-time predictions
- Personalized AI advice
- Professional report generation
- User-friendly interface

3

Try It Yourself

Google Collab : Link

Thank You

Acknowledgements

- UCI for the dataset
- GroqCloud for AI API
- Open-source community
- Course instructors